

# Project Executable Files

|               |                                            |
|---------------|--------------------------------------------|
| Date          | 15 July 2026                               |
| Team ID       | XXXXXX                                     |
| Project Name  | FloodGuard - AI/ML Flood Prediction System |
| Maximum Marks | 3 Marks                                    |

## Step 1: Submission Checklist

| S.No | Item to Submit                                           | Submitted (Yes/No/NA)                           |
|------|----------------------------------------------------------|-------------------------------------------------|
| 1    | Complete source code (all files and folders)             | Yes                                             |
| 2    | README / Setup Guide (instructions to run the project)   | Yes                                             |
| 3    | requirements.txt / dependency file                       | Yes                                             |
| 4    | Database schema / seed files (if applicable)             | Yes (app/models.py, app/seed.py)                |
| 5    | Environment configuration file (.env.example or similar) | NA (no secrets required for base functionality) |
| 6    | Deployed application URL (if hosted)                     | Yes                                             |
| 7    | APK / Executable binary (if applicable)                  | NA (web application)                            |
| 8    | Dockerfile / Containerization config (if applicable)     | NA (deployed directly via Render buildpacks)    |
| 9    | Test files and test results                              | Yes (Performance Testing document)              |
| 10   | Demo video or walkthrough (if required)                  | To be added                                     |

## Step 2: File / Folder Structure

### Project Structure

```
flood-prediction-system-main/
├── main.py
├── train.py
├── requirements.txt
├── app/
│   ├── __init__.py
│   ├── models.py
│   ├── ml.py
│   ├── routes.py
│   └── seed.py
├── templates/
│   ├── base.html
│   ├── home.html
│   ├── predict.html
│   ├── history.html
│   ├── 404.html
│   └── 500.html
├── static/
│   ├── css/style.css
│   ├── js/script.js
│   └── img/
│       ├── class_distribution.png
│       ├── correlation_heatmap.png
│       ├── model_comparison.png
│       └── best_model_confusion_matrix.png
├── models/
│   ├── flood_model.joblib
│   ├── scaler.joblib
│   └── metadata.json
└── data/
    └── flood_dataset.xlsx
```

### Step 3: Deployment / Access Details

| Field                       | Details                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL       | <a href="https://floodguard-ytqr.onrender.com">https://floodguard-ytqr.onrender.com</a>                             |
| Login Credentials (Demo)    | Not required — the prediction form is publicly accessible                                                           |
| Platform / Hosting Provider | Render (Web Service, Free Plan)                                                                                     |
| Repository Link             | <a href="https://github.com/BSG8710/flood-prediction-system">https://github.com/BSG8710/flood-prediction-system</a> |
| Demo Video Link             | To be added                                                                                                         |

### Step 4: Run Instructions

- Clone the repository: `git clone https://github.com/BSG8710/flood-prediction-system`
- Create and activate a virtual environment (`python -m venv venv`)
- Install dependencies: `pip install -r requirements.txt`
- Train the model: `python train.py`
- Run the app: `python app.py`
- Open `http://localhost:5000` in a browser

### Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation                                         | Workaround / Status                                          |
|------|------------------------------------------------------------------|--------------------------------------------------------------|
| 1    | Render free tier spins down after 15 min idle (~60s cold start)  | Acceptable for demo use; upgrade to paid tier to remove      |
| 2    | Prediction history resets on redeploy (SQLite on ephemeral disk) | Planned: migrate to Render's free PostgreSQL for persistence |
| 3    | Small training dataset (115 rows, imbalanced)                    | Mitigated with class-weighting and F1-based model selection  |